MIL-S-46047E(MR)
21 September 1988
SUPERSEDING
MIL-S-46047D(MR)
1 May 1978

MILITARY SPECIFICATION

STEEL, BARS AND BLANKS (1-3/4 INCHES AND OVER IN DIAMETER) FOR BARRELS OF SMALL ARMS WEAPONS

This specification is approved for use by the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel alloy bars and blanks 1-3/4 inches and over in diameter of one chemical composition for use in the manufacture of barrels for small arms weapons (see 6.1).

1.2 Classification. Bars and blanks shall be furnished in the following conditions, as specified (see 6.3):

Bars (Mill length).
Hot-rolled, as-rolled
Hot-rolled and annealed to a maximum Brinell hardness of HB250 for cold sawing.

Blanks or gun barrel bar lengths.
As-rolled (rolled, forged, extruded, or upset).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MEE, Watertown, MA 02172-0001 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9510

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-8697 - Electrodes, Welding, Coated, Low-Hydrogen, Heat-Treatable Steel

STANDARDS

FEDERAL

FED-STD-48 - Tolerances for Steel and Iron Wrought Products

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM

ASTM A255 - Method of End Quench Test for Hardenability of Steel
ASTM A751 - Chemical Analysis of Steel Products
ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E23 - Notched Bar Impact Testing of Metallic Materials
ASTM E381 - Macroetch Testing, Inspection, and Rating Steel Products, Comprising Bars, Billets, Blooms, and Forgings

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

AMS 2640 - Magnetic Particle Inspection

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements shown in table I.

TABLE I. Chemical composition.1/

Element	Ladle analysis2/	Check analysis3/
	Percent	Percent
Carbon	0.38-0.45	0.36-0.47
Manganese	.75-1.00	.71-1.04
Phosphorus	.025	.030
Sulfur	.020	.025
Silicon	.20- .35	.18- .37
Chromium	.95-1.25	.90-1.30
Molybdenum	.55- .70	.52- .73
Vanadium	.20- .30	.17- .33

1/ Chemical ranges and limits based on ladle analysis.

2/ Steel containing elements not designated, in excess of the following amounts, shall be subject to rejection: Copper 0.35 percent and aluminum 0.040 percent.

3/ Maximum except where indicated as a range.

3.1.1 Ladle analysis. A certified ladle analysis of each heat or melt of steel shall be furnished by the contractor showing the percentages of the elements designated.

3.1.2 Check analysis. The chemical composition, as determined by check analysis, shall meet the applicable requirements specified in table I when tested as specified in 4.4.1 and 4.6.1.

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3.2 Impact resistance. Each pair of test specimens from each sample taken as specified in 4.4.2 and heat-treated as specified in 4.6.2 to a Rockwell hardness of not less than 35 HRC shall have an average transverse impact value at a temperature of minus 40°F + 2°F of not less than 10 foot-pounds when tested as specified in 4.6.2. No individual value shall be less than 7 foot-pounds.

3.3 Macrostructure. The quality and cleanliness of the steel as indicated by the results of the macroetch test specified in 4.4.3 and 4.6.3 shall be equal to or better than macrographs S-2, R-2, and C-2 for electric furnace process and S-3, R-3, and C-3 for open hearth process of ASTM E381.

3.4 Nonmetallic inclusions. The steel shall have a maximum average rating of 0.45 for both frequency and severity when tested as specified in 4.4.4, 4.6.4, and 4.6.5.1. No individual test specimen shall have a severity rating greater than 0.75.

3.5 Hardenability. The hardenability of the steel shall be not less than 54 HRC at J 20/16 inches (1 1/4 inches from the end of standard Jominy or end-quench test specimen) when tested as specified in 4.4.5 and 4.6.5.

3.6 Forging practice. When forging operations are performed, process controls shall be subject to Government approval. Heating for forming shall be conducted using methods and equipment, including pyrometric controls, suitable for the purpose.

3.7 Dimensions and tolerances.

3.7.1 Blanks and forged, extruded, or upset bars. Blanks and forged, extruded, or upset bars shall conform to the dimensions and tolerances specified in the contract or order (see 6.3).

3.7.2 Hot-rolled bars. Hot-rolled bars shall conform to the ordered dimensions and shall be within the applicable tolerances shown in FED-STD-48.

3.8 Marking. Each bar or blank shall be suitably marked to identify the heat or melt. The marking medium shall not react objectionably with the surface of the steel during heat treatment. When specified, individual bars or blanks shall be identified as required in the contract or order (see 6.3).

3.9 Workmanship. Bars and blanks shall be sound, uniform in quality and condition; and free of injurious laps, cracks, indications of overheating or burning, twists, seams, damaged ends, and other related defects injurious to the finished component.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. All examination and testing in this specification is classified as quality conformance inspection and shall be to determine conformance to the requirements of the specification to serve as a basis for acceptance of the material covered by this specification.

4.3 Lot. A lot shall consist of all material from the same heat or melt and of the same condition and size submitted for inspection at one time. Identification of each heat or melt shall be maintained throughout manufacture and inspection.

4.4 Sampling.

4.4.1 For chemical analysis. At least one sample for chemical analysis shall be taken from each lot for chemical analysis.

4.4.2 For impact resistance testing. One sample shall be taken from material representing the top and bottom of the first and last usable ingots of the heat or melt (four samples per heat or melt.) At the option of the Government, random sampling may be used by taking at least 1 bar or blank for each 10,000 pounds of material, or fraction thereof. Blooms and billets may be forged to bar or blank diameter for use as samples for this testing but in any case the sample to be tested shall be of diameter specified in the contract or order (see 6.3).

4.4.3 For macroetch testing. Full cross-section samples shall be taken, after discard, from billets representing the top and bottom of the first, middle, and last usable ingots of each heat or melt. At the option of the Government, random sampling may be used by taking a full cross-section sample from at least 1 billet in each 10,000 pounds of material, or fraction thereof. In the absence of billet tests, 10 bars shall be taken from each lot.

4.4.4 For nonmetallic inclusion testing. A sample shall be taken from bars and blanks representing the top and bottom of the first and last usable ingots from heats or melts having not over 10 ingots or more than 30 tons, or from portions of heats or melts within these limits; and from the top and bottom of the first, middle, and last usable ingots of heats or melts having more than 10 ingots or over 30 tons. At the option of the Government, random sampling may be used by taking a sample of 10 bars or blanks from each lot. Blooms and billets may be forged to finish bar or blank size for use as samples for this testing. When specified in the contract or order (see 6.3), the contractor

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shall submit to the procuring agency, prior to or not later than delivery of the product of the heat or melt, test samples from the same bars or blanks from which the contractor's test samples were taken. These samples shall be approximately 14 inches long (see 4.6.4).

4.4.5 For hardenability testing. At least one sample shall be taken from material representing the top and bottom of the first and last usable ingots in the heat or melt (four samples per heat or melt).

4.5 Examination.

4.5.1 Forging practice. Periodic surveillance of forging practice shall be conducted to assure compliance with the requirements of 3.6. The government representative shall have access to pyrometers and recording instruments and to contractor records. The accuracy of the pyrometers and sensing devices shall be checked by the contractor using suitable calibrating equipment whenever requested by the Government representative.

4.5.2 Visual. Unless otherwise specified in the contract or order, all material in each lot shall be subject to visual examination to insure conformance with the requirements for marking (see 3.8) and for workmanship (see 3.9).

4.5.3 Dimensions and tolerances. Unless otherwise specified in the contract or order, a represenative number of bars from each lot shall be measured to determine compliance with the requirement for dimensions and tolerances (see 3.7).

4.5.4 Preparation for shipment. Examination of the preservation, packaging, packing, and marking for shipment shall be made for conformance to the requirements of section 5.

4.6 Tests.

4.6.1 Chemical analysis. Chemical analysis of the samples (see 4.4.1) shall be conducted in accordance with ASTM A751.

4.6.2 Impact tests. Sample bars or blanks taken in accordance with 4.4.2 or cylindrical test coupons taken from a sample bar shall be heat treated by quenching in circulating oil after austenitizing a minimum of 90 minutes at a temperature of $1600^{\circ} \pm 25^{\circ}F$ followed by uniform tempering of the hardness specified in 3.2 as employed in the production of finished items. Preliminary normalizing may be used. After heat treatment, material sufficient to prepare at least two transverse V-notch Charpy impact specimens shall be taken from each bar, blank, or test coupon so that the longitudinal dimensions of the specimen are in the plane of, and so that the bottom of the notch is approximately tangent to, a circle, the circumference of which is the inside surface of the finished gun tube bore, as illustrated in figure 1 (see 6.3). Test specimens shall be prepared and tested in accordance with ASTM E23. The temperature of the test specimens shall be $-40^{\circ} \pm 2^{\circ}F$. Charpy impact test machines shall have been calibrated within one year of the contract. If, because of insufficient material, the corners opposite the notched sides of the specimens cannot be finished, test results from such specimens shall be accepted. When the diameter of the finished bar or blank does not permit a

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specimen with a notched side of standard length (2.165 inch) to be made, pads shall be welded to its periphery, prior to heat treatment, as necessary to permit machining of standard specimen except that the length opposite the notched face may be less than 2.165 inches. Welding shall be performed using the covered electrode arc process or the inert-gas shielded electric arc process. A class HT-4130 electrode conforming to MIL-E-8697 shall be used with the covered electrode process, and a bare wire electrode 4130 steel shall be used with the inert-gas shielded process. Hardness readings to determine compliance with 3.2 shall be taken in accordance with ASTM E18 on the notched face of the hardened test specimen within one-fourth inch of the center line of the notch. The error in the Charpy machine shall not exceed ±1 ft./lb. for energy values up to 20 ft./lb. or ±5 percent for energy values over 20 ft./lb. when machine is tested using comparison specimens which may be obtained from the procuring activity (see 6.2).

4.6.3 Macrostructure. Test specimens shall be prepared and tested in accordance with ASTM E381.

4.6.4 Nonmetallic inclusions. Unless otherwise specified, testing for nonmetallic inclusions shall be conducted using the step-down specimen. When specified in the contract or order (see 6.3), a straight cylindrical specimen may be used in lieu of the step-down specimen.

4.6.4.1 Step-down specimen. The step-down specimen shall be generated in equal length circumferential steps as follows:

Bar or blank diameter	Step length	Step diameter				
Inches	Inches	1	2	3	4	5
2.000 to 3.500 incl.	1.000	D	4/5D	3/5D	2/5D	1/5D

4.6.4.2 Straight cylindrical specimen. The specimen shall be machined in accordance with the machining allowance specified in 4.6.4.3, to a straight cylindrical specimen 5 inches ± 1/16 inch in length.

4.6.4.3 Stock removal.

Diameter	Minimum stock removal1/
Inches	Inch (measured on radius)
2,000	0.090
Over 2,000 to 2,500 incl.	0.125
Over 2,500 to 3,500 incl.	0.156

1/ Allow 0.010 inch for finish machining after heat treatment

4.6.4.4 Heat treatment. Rough machined specimens shall be hardened by suitably austenitizing, quenching, and tempering to produce a Brinell hardness of HB 301 to 344.

4.6.4.5 Surface finish. The finish machined surface of the test specimen shall not exceed a roughness height rating of 40 micro-inches. The ends shall be finished to provide good electrical contact.

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4.6.5 Hardenability. Test method ASTM A255 shall be used in the preparation of the specimens and in the hardenability testing except that the austenitizing temperature shall be 1600° ± 25°F.

4.6.5.1 Testing. Testing for non-metallic inclusions shall be by the wet magnetic particle method specified in AMS 2640 using circular magnetization with a magnetizing current of 900 to 1,100 amperes per inch of diameter. Direct current shall be used for magnetizing. When the step-down specimen is used, the smallest step shall be magnetized and inspected first; the next larger steps shall be magnetized and inspected in succession. Inclusions one-eighth inch or less in length shall not be counted. The frequency and severity of nonmetallic inclusions shall be determined as shown below:

Frequency

a. Total the number of indications for each specimen.
b. Divide the total number of indications per specimen by the surface area of the specimen in square inches. This value is the frequency rating of the specimen.
c. Total the frequency ratings of the specimens.
d. Divide the total frequency ratings by the number of specimens. This value is the average frequency.

Severity

a. The length of each indication is recorded.
b. Total the number of indications for each size interval shown in table II.
c. Multiply the total number per size interval by the severity factor of table II.
d. Total the products obtained in (c) for each specimen.
e. Divide the product total by the surface area of the specimen in square inches. This value is the severity rating for the specimen.
f. Total the severity ratings of the specimens.
g. Divide the total severity ratings by the number of specimens. This value is the average severity rating.

TABLE II. Severity of nonmetallic inclusions.

Indication length[1/]	Severity factor
Inch	
Over 1/8 to 1/4 incl.	1
Over 1/4 to 1/2 incl.	2
Over 1/2 to 3/4 incl.	4
Over 3/4 to 1 incl.	8
Over 1	16

1/ When the step-down specimen is used, an inclusion extending the entire length of a step shall be considered to be longer than 1 inch and shall be assigned a weight of 16.

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4.7 Rejection and retest. Unless otherwise specified in the contract or order, rejection and retest shall be conducted in accordance with ASTM A751.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.3).

5.1.1 Level A. Cleaning, drying, preservation, and packaging shall be in accordance with MIL-STD-163.

5.1.2 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with manufacturer's commercial practice.

5.2 Packing. Packing shall be level A or level C as specified (see 6.3).

5.2.1 Level A. Packing shall be in accordance with MIL-STD-163.

5.2.2 Level C. Material shall be packed in a manner to ensure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Steel covered by this specification is intended for use in the manufacture of gun barrels requiring deep hole drilling and rifling operations.

6.1.1 Dimensional limitation. Dimensional limitation set by the scope of this specification applies to the outside diameter of the bars and blanks as heat treated. Application of this specification to bars and blanks exceeding 2-5/8 inches in diameter should not be made without review of the responsible technical agency and suitable modification of chemistry and hardenability.

6.1.2 Charpy specimen. Charpy specimens used for comparison tests should be obtained from the local DSA(DCSAR) office or, in cases where DCASR does not have responsibility over a particular contract, from the U.S. Army Materials Technology Laboratory, Watertown, MA 02172-0001.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether bars or blanks are required and the condition required (see 1.2).

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c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Size required (see 3.7).
e. Special marking required on bars or blanks (see 3.8).
f. Diameter of sample to be tested for impact resistance (see 4.4.2).
g. Whether test samples are required by procuring agency (see 4.4.4).
h. When changes in visual and dimensional examination requirements are permissible (see 4.5.2 and 4.5.3).
i. Gun tube bore diameter (see 4.6.2).
j. When straight cylindrical specimen is specified for nonmetallic inclusions testing (see 4.6.4).
k. Selection of applicable levels of preservation, packaging, and packing (see section 5).
l. When special marking is required (see 5.3).

6.3 Metric units. When required the units for inch, foot, pound, pounds per square inch, and feet per second may be converted to the metric equivalents by multiplying them by the following conversion factors:

English unit	Multiplied by	Equals	Metric equivalent
Inch	0.025400	=	Metre (m)
Foot	0.304800	=	Metre (m)
Pounds (lb)	0.453592	=	Kilograms (kg)
Pounds per square inch (psi)	0.006895	=	Mega Pascal (MPa)
Feet per second (FT/s)	0.304800	=	Metre per seconds, (m/s)

Note: Conversion factors can be associated with ASTM E380 entitled "Metric Practice Guide."

6.4 Subject term (key word) listing.

Bars and Blanks
Gun Barrels
Small Arms Weapons
Steel Alloy

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - AR
DLA - IS

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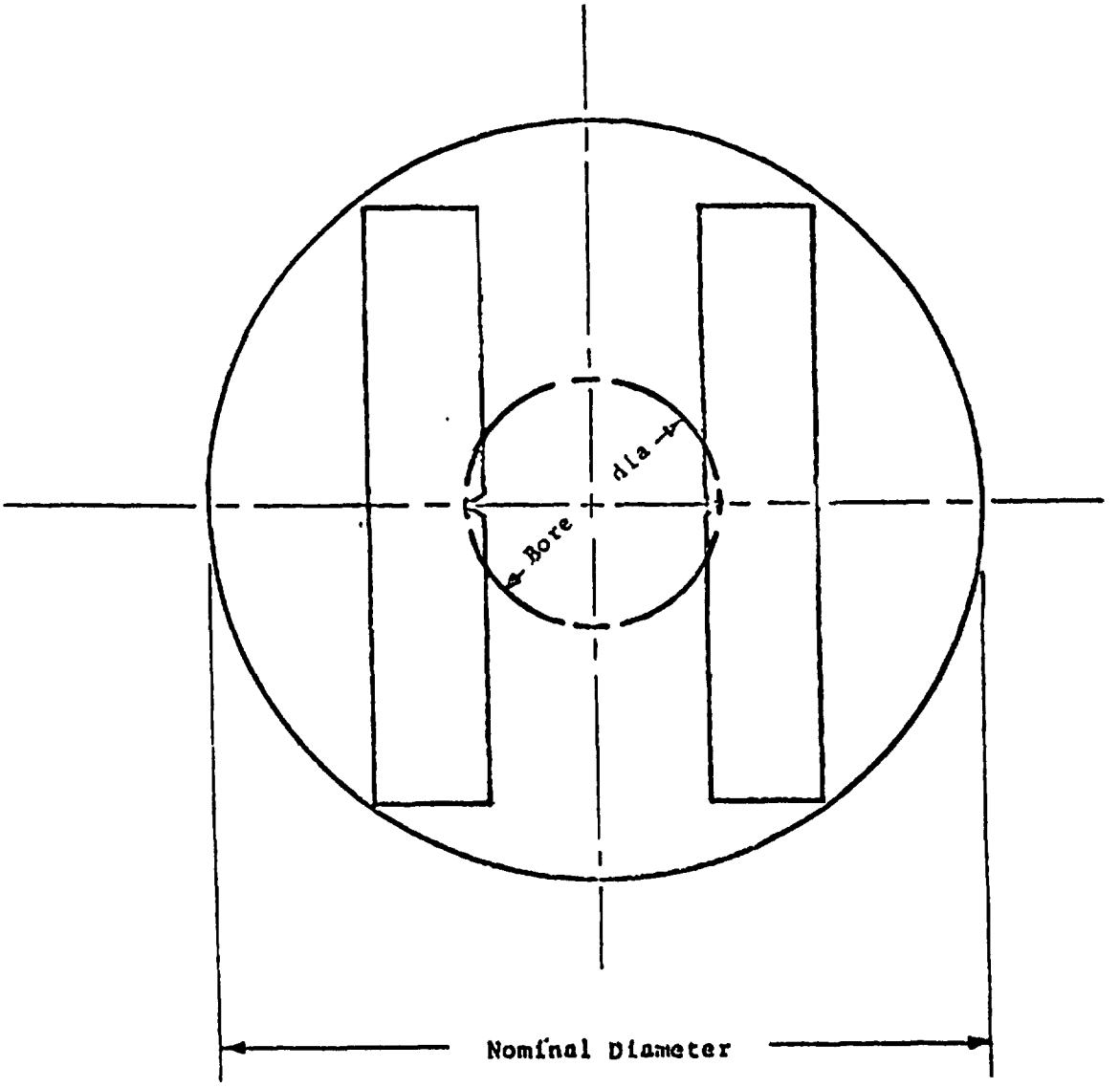


FIGURE 1. Location of transverse V-Notch Charpy impact specimens in test sample.

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1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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